Linear Algebra Lecture Notes

Chapter 6

6.1 Inner Product, length and orthogonality

– If u and v are vectors in $\mathbb{R}^n$, then we regard u and v as $n \times 1$ matrices. The traspose u^T is a $1 \times n$ matrix, and the matrix product $u^T v$ is a 1×1 matrix, which we write as a single real number (a scalar) without brackets. The number $u^T v$ is called the inner product of u and v, and is often written as $u \cdot v$. This inner product is also referred to as a dot product. If

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

then the inner product of u and v is

$$\begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n .$$

Ex: Compute $u \cdot v$ and $v \cdot u$ for $u = \begin{bmatrix} 2 \\ -5 \\ -1 \end{bmatrix}$ and $v = \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix}$

Sol

$$u \cdot v = u^T v = \begin{bmatrix} 2 & -5 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix} = 2 \cdot 3 + (-5)\cdot 2 + (-1)\cdot(-3) = -1$$

$$v \cdot u = v^T u = \begin{bmatrix} 3 & 2 & -3 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \\ -1 \end{bmatrix} = 3 \cdot 2 + (2)\cdot(-5) + (-3)\cdot(-1) = -1$$

– It is clear from the calculations in the above example why $u \cdot v = v \cdot u$.

Theorem:

Let **u**, **v**, and **w** be vectors in $\mathbb{R}^n$, and let c be a scalar. Then

a. $\mathbf{u}\cdot\mathbf{v} = \mathbf{v}\cdot\mathbf{u}$ ✓

b. $(\mathbf{u}+\mathbf{v})\cdot\mathbf{w} = \mathbf{u}\cdot\mathbf{w} + \mathbf{v}\cdot\mathbf{w}$ ✓

c. $(c\mathbf{u})\cdot\mathbf{v} = c(\mathbf{u}\cdot\mathbf{v}) = \mathbf{u}\cdot(c\mathbf{v})$ ✓

d. $\mathbf{u}\cdot\mathbf{u} \geq 0$, and $\mathbf{u}\cdot\mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$ ✓

$u = [a \;\; b \;\; c]$

$u \cdot u = a^2 + b^2 + c^2 = 0$

only when $a = b = c = 0$

i.e $u = 0$

– Properties (b) and (c) can be combined several times to produce the following useful fact:

$$(c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p)\cdot\mathbf{w} = c_1(\mathbf{u}_1\cdot\mathbf{w}) + \cdots + c_p(\mathbf{u}_p\cdot\mathbf{w})$$

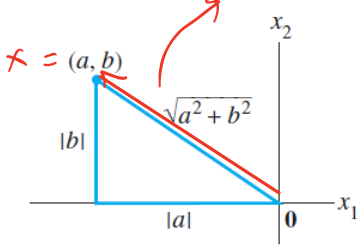
Definition:

The **length** (or **norm**) of **v** is the nonnegative scalar $\|\mathbf{v}\|$ defined by

$$\|\mathbf{v}\| = \sqrt{\mathbf{v}\cdot\mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}, \quad \text{and} \quad \|\mathbf{v}\|^2 = \mathbf{v}\cdot\mathbf{v}$$

– Suppose v is in $\mathbb{R}^2$, say $v = \begin{bmatrix} a \\ b \end{bmatrix}$. If we identify v with a geometric point in the plane, then $\|v\|$ coincides with the standard notion of the length of the line segment from the origin to v. This follows from the Pythagorean Theorem applied to a triangle such as the one in the following figure.

$\|x\|$ = length of x



– A similar calculation with the diagonal of a rectangular box shows that the definition of length of a vector in $\mathbb{R}^3$ coincides with the usual notion of length.

– For any scalar c, the length of cv is $|c|$ times the length of v. That is,

$$\|c\boldsymbol{v}\| = |c|\|\boldsymbol{v}\|$$

Compute $\|cv\|^2 = (cv)\cdot(cv) = c^2 v\cdot v = c^2\|v\|^2$ and take the roots.

– A vector whose length is 1 is called a unit vector. If we divide a nonzero vector $\boldsymbol{v}$ by its length – that is, multiply by $1/\|\boldsymbol{v}\|$ – we obtain a unit vector $\boldsymbol{u}$ because the length of $\boldsymbol{u}$ is $(1/\|\boldsymbol{v}\|)\|\boldsymbol{v}\|$. The process of creating $\boldsymbol{u}$ from $\boldsymbol{v}$ is sometimes called normalizing $\boldsymbol{v}$, and we say that $\boldsymbol{u}$ is in the same direction as $\boldsymbol{v}$.

Ex: let $v = \begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix}$. Find a unit vector u in the same direction as v.

sol Compute length of v:

$$\|v\|^2 = v\cdot v = (1)^2 + (-2)^2 + (2)^2 + (0)^2 = 9$$

$$\|v\| = \sqrt{9} = 3$$

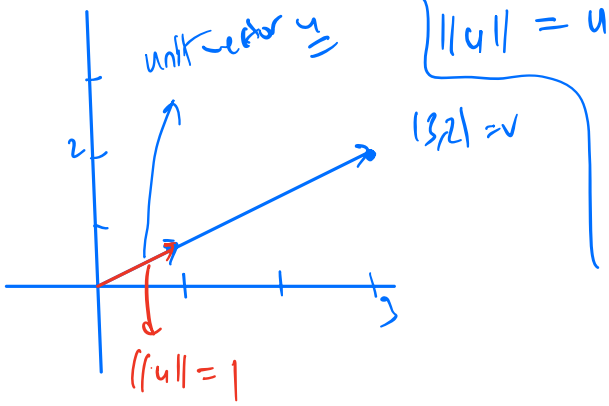
Multiply v by $1/\|v\|$:

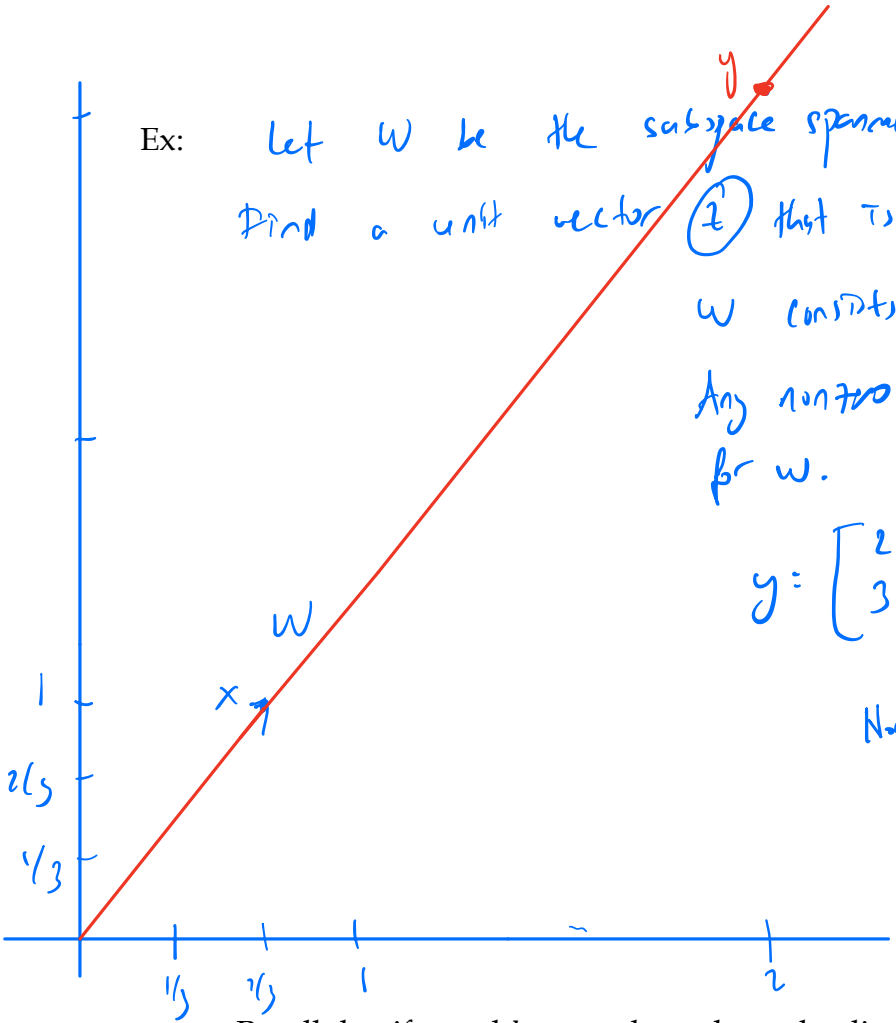
$$\frac{1}{\|v\|}v = \frac{1}{3}v = \frac{1}{3}\begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \\ 0 \end{bmatrix} = u$$

Check and see that

$$\|u\|^2 = u\cdot u = \left(\frac{1}{3}\right)^2 + \left(\frac{-2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + (0)^2$$

$$= \frac{1}{9} + \frac{4}{9} + \frac{4}{9} + 0 = \frac{9}{9} = 1 \;//$$





Ex: Let W be the subspace spanned by $x = \left(\frac{2}{3}, 1\right)$. Find a unit vector z that is a basis for W.

W consists of all multiples of x.

Any nonzero vector in W is a basis for W. Scale x to make calculations easy

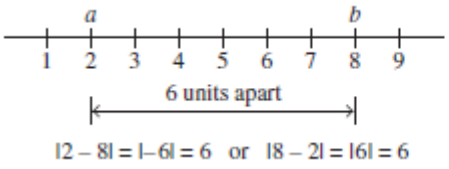
$y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $\quad \|y\|^2 = y \cdot y = 2^2 + 3^2 = 13$

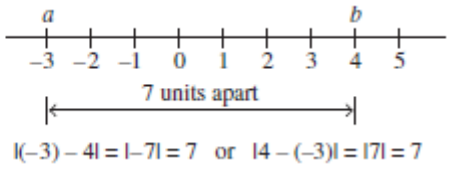
$\|y\| = \sqrt{13}$

Normalize y:

$z = \frac{1}{\sqrt{13}} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{13} \\ 3/\sqrt{13} \end{bmatrix}$

– Recall that if a and b are real numbers, the distance on the number line between a and b is the number $|a - b|$. See the following figure. This definition of distance in $\mathbb{R}$ has a direct analogue in $\mathbb{R}^n$.





Definition:

For $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, the **distance between $\mathbf{u}$ and $\mathbf{v}$**, written as $\text{dist}(\mathbf{u}, \mathbf{v})$, is the length of the vector $\mathbf{u} - \mathbf{v}$. That is,

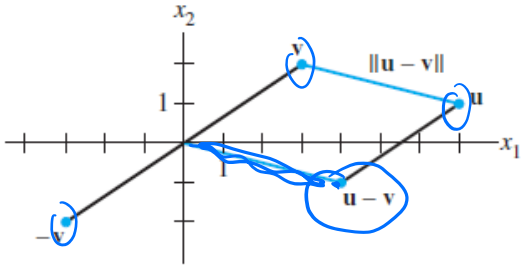
$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Ex: $u = (7,1)$ $v = (3,2)$ Calculate $dist(u,v)$

Sol

$$u - v = \begin{bmatrix} 7 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$\|u - v\| = \sqrt{4^2 + (-1)^2} = \sqrt{17}$$



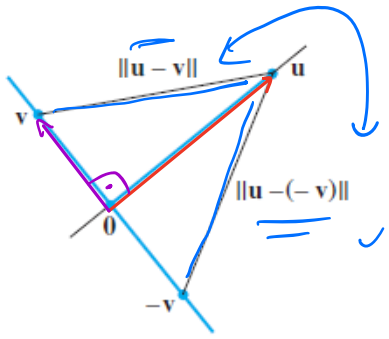
Ex: If $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$, then

$$dist(u,v) = \|u - v\| = \sqrt{(u-v)\cdot(u-v)}$$

$$= \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2}$$

Orthogonal Vectors

- Consider $\mathbb{R}^2$ or $\mathbb{R}^3$ and two lines through the origin determined by vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ as shown in the next figure.



- The two lines are geometrically perpendicular if and only if the distance from $\boldsymbol{u}$ to $\boldsymbol{v}$ is the same as the distance from $\boldsymbol{u}$ to $-\boldsymbol{v}$. This the same as requiring the squares of the distances to be the same. Now …

$$(dist(u,-v))^2 = \|u-(-v)\|^2 = \|u+v\|^2$$
$$= (u+v)\cdot(u+v)$$
$$= u\cdot(u+v) + v\cdot(u+v)$$
$$= u\cdot u + u\cdot v + v\cdot u + v\cdot v$$
$$= \|u\|^2 + \|v\|^2 + 2u\cdot v$$

$$(dist(u,v))^2 = \|u-v\|^2$$
$$= (u-v)\cdot(u-v)$$
$$= u\cdot(u-v) - v\cdot(u-v)$$
$$= u\cdot u - u\cdot v - v\cdot u + v\cdot v$$
$$= \|u\|^2 + \|v\|^2 - 2v\cdot u$$

$$(dist(u,-v))^2 = (dist(u,v))^2 = \|u\|^2 + \|v\|^2 + 2u\cdot v = \|u\|^2 + \|v\|^2 - 2u\cdot v$$

Two squared distances are equal if and only if $2u \cdot v = -2u \cdot v$ which happens if and only if $u \cdot v = 0$.

– This calculation shows that when vectors u and v are identified with geometric points, the corresponding lines through the points and the origin are perpendicular if and only if $\boldsymbol{u} \cdot \boldsymbol{v} = 0$. The following definition generalizes to $\mathbb{R}^n$ this notion of perpendicularity (or orthogonality).

Definition:

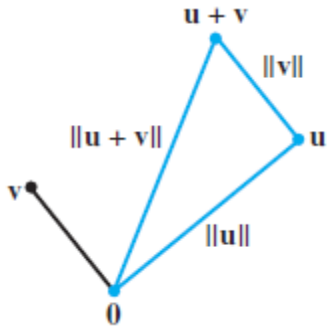
Two vectors **u** and **v** in $\mathbb{R}^n$ are **orthogonal** (to each other) if $\mathbf{u} \cdot \mathbf{v} = 0$.

– Observe that the zero vector is orthogonal to every vector in $\mathbb{R}^n$ because $\mathbf{0}^T \boldsymbol{v} = 0$.

Theorem:

The Pythagorean Theorem

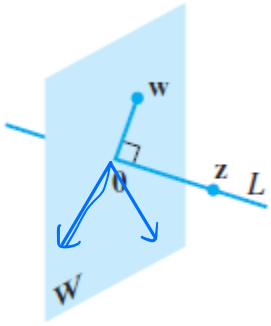
Two vectors **u** and **v** are orthogonal if and only if $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.



– If a vector z is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then z is said to be orthogonal to W.

– The set of all vectors z that are orthogonal to W is called the orthogonal complement of W and is denoted by $W^{\perp}$ (read as "W perpendicular" or "W perp").

Ex:



$z \cdot w = 0$

So each vector on L is orthogonal to every w in W. In fact, L consist of all vectors that are orthogonal to the w's in W, and W consists of all vectors orthogonal to z's in L.

That is $L = W^{\perp}$ and $W = L^{\perp}$